Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (EE) Examination

March-April-2018

EE 243 Electrical Measurements and Industrial Instrumentation

Date: 02.04.2018, Monday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 What is synchronizing? Describe under which conditions a 3 phase alternator can be synchronized to 3 bus bars? **[07]**

Q – 2.a Draw the phasor diagram and equivalent circuit of Current transformer **[04]**

Q – 2.b **Answer any two questions.** **[10]**

a) Describe any one method for testing of a current transformer.

b) Discuss in short about Strip Chart Recorders.

c) Explain the difference between sensor, transducer and actuator with example.

Q - 3 **Answer any two.** **[14]**

a) Explain the construction and working of the Weston type frequency meter.

b) Draw the block diagram of DMM (digital multimeter) and explain each part in short.

c) Draw the phasor diagram of CT and derive the equation of transformation ratio and phase angle error for CT

SECTION – II

Q - 4 .a **Indicate the correct statement in the following** **[04]**

1 Which of the following thermocouples will give the highest output for the same value of hot and cold junction temperature?

(a) Iron-Constantan (b) Chromel- Constantan

(c) Platinum-cadmium (d) All of above

2 The deflection of the free end of the bimetallic strips in a bimetallic thermometer with temperature is nearly

(a) Linear (b) Non-Linear

(c) Hyperbolic (d) Parabolic

3 Function of transducer is to convert

(a)	Electrical signal into non electrical quantity	(b)	Non electrical quantity into electrical signal
(c)	Electrical signal into mechanical quantity	(d)	All of these

4 Which metal has the most linear resistance temperature characteristics?

(a)	Copper	(b)	Nickel
(c)	Platinum	(d)	Cadmium

Q - 4 .b Describe in short about capillary tube viscometer. **[03]**

Q – 5.a Explain McLeod Gauge in detail. **[04]**

Q – 5.b **Answer any two questions.** **[10]**

a) Explain the bubbler system for specific gravity measurement.

b) Explain the Dip-stick method for level measurement.

c) Explain bourdon tube in detail with neat and clean diagram.

d) Describe in short about the Gas chromatography.

Q - 6 **Answer any two.** **[14]**

a) Explain in detail about nuclear absorption method for density measurement.

b) Explain moving magnet type velocity transducer.

c) Write about the Orifice plate in detail.

d) Explain Infrared radiation pyrometer in detail.

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